

CS & IT ENGINEERING



Operating System

Process Management

Lecture -2

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Recap of Previous Lecture



Topic

Dual Mode of Operation

Topic

Process

Topic

Process Representation

Topic

Process Control Block

Topics to be Covered



Topic

Process states

Topic

Process State Transition

Topic

Process Scheduling

Topic

Types of Schedulers

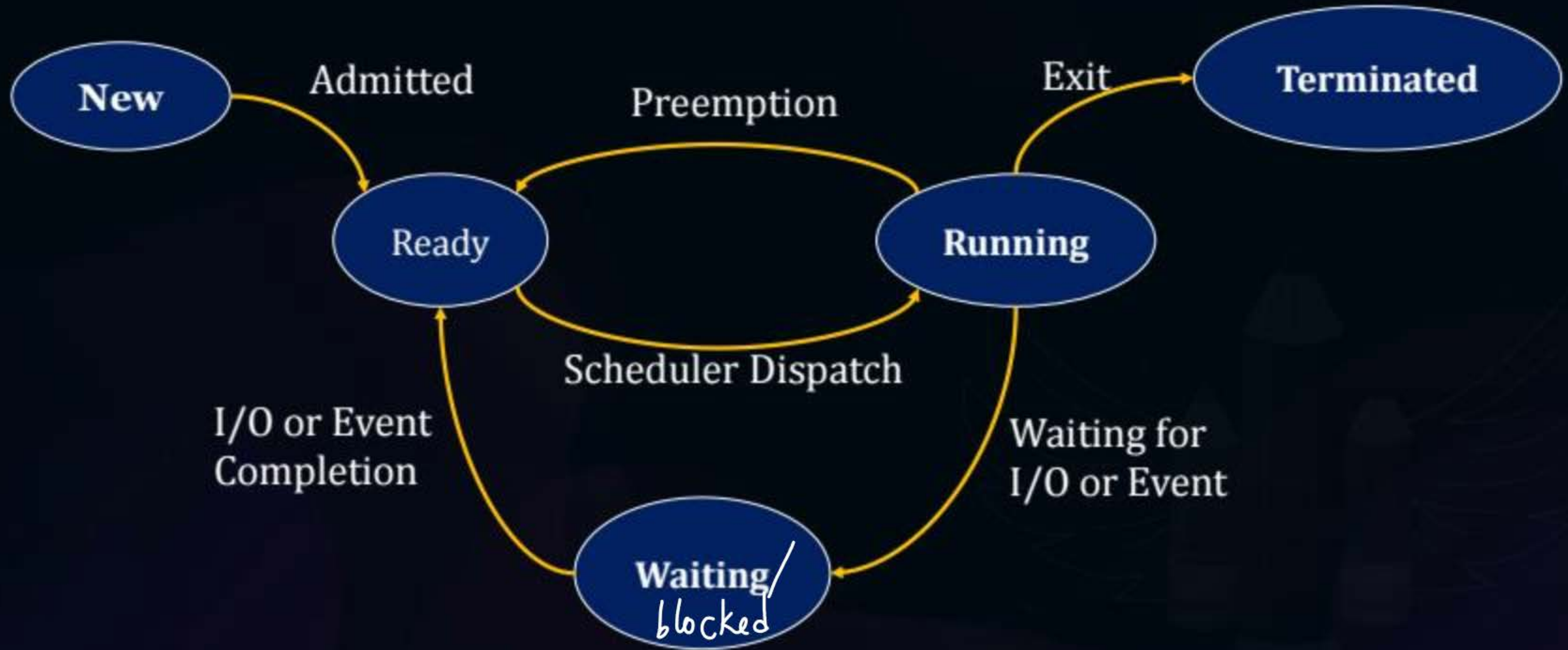


Topic : Process States

↓
what activity process is currently doing .



Topic : Process States





Topic : Process States

(but not running)

New:- All installed processes are known to be in new state

Ready:- All processes which are waiting to run on CPU are known to be in ready state

Running:- A process which is running on CPU has its state as running

Terminated:- A completed process has its state as terminated

Blocked:- All processes which are waiting for any I/O or event



Topic : Process States



New To Ready: When process is admitted by OS (resource allocation mem. etc.)

Ready to Running: When a process is dispatched to CPU

Running to Terminated: When a process is completed (resource deallocated)

Running to Blocked: When a process goes for IO or event

Running to Ready: When a process is preempted

Blocked to Ready: When a process completes IO or event

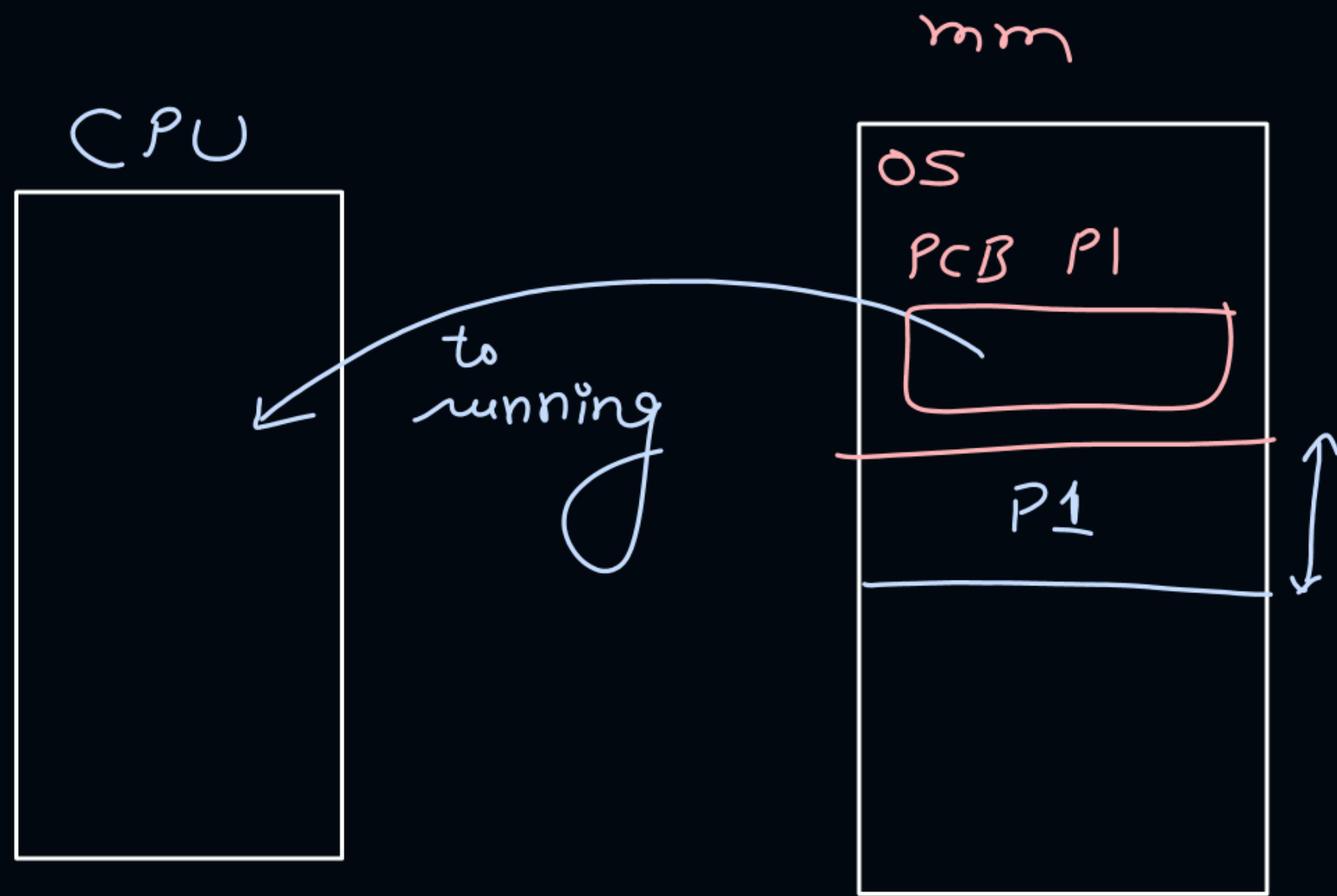


Topic : Process States



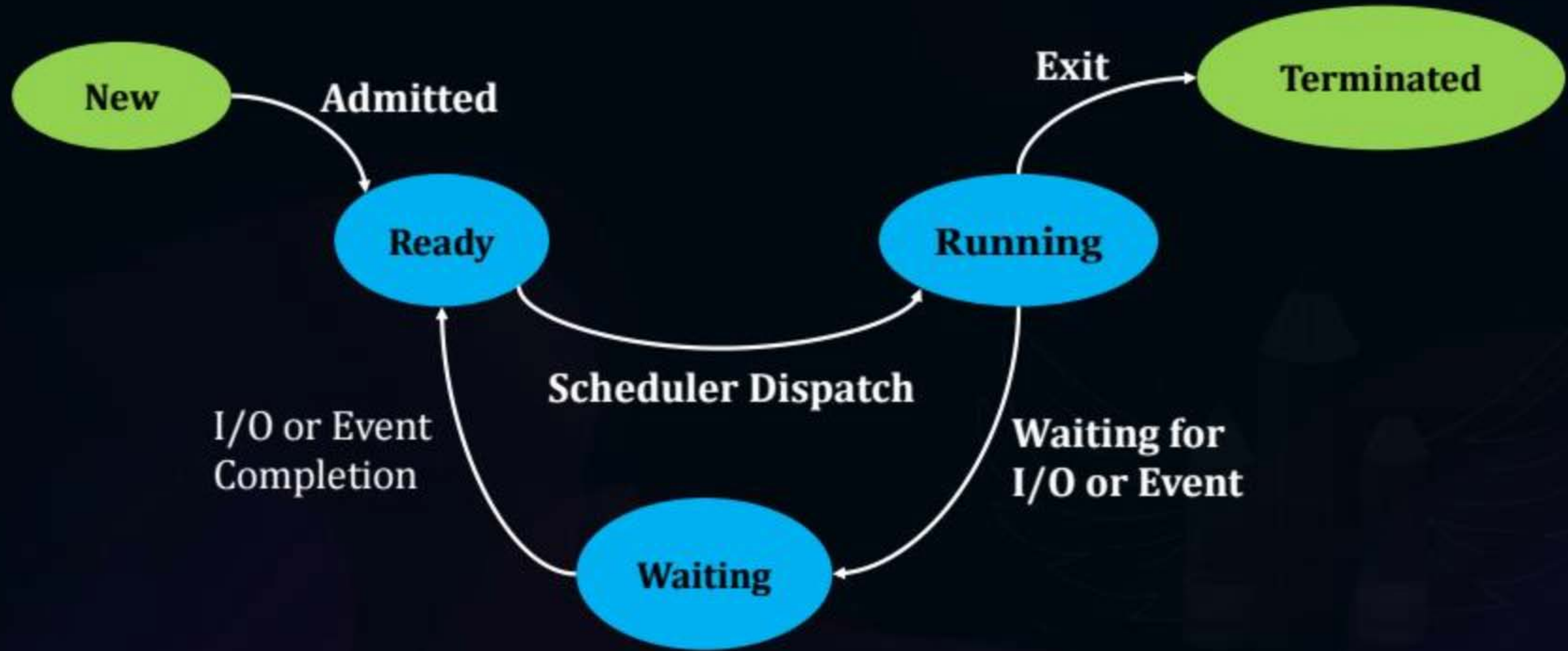
- ❑ 2 Transitions are voluntary → Process can initiate by itself.
 - Running to Terminated
 - Running to Blocked

A process is in mm when it is in ready, running or blocked.





Topic : Non-preemptive Process States





Topic : CPU vs IO Bound Process

CPU intensive

CPU Bound: If the process is intensive in terms of CPU operations

IO Bound: If the process is intensive in terms of IO operations

IO intensive



Topic : Process Scheduling

Needed Because?

↓
{ in multiprogramming system
to run processes on CPU & to select which
process will use which I/O device at what time.

Goal:- efficiency or better performance



Topic : Scheduling Queues

- Job Queue → all the new state processes are kept in it.
 - Ready Queue → all the ready — || ——— || ———
 - Device Queue
↳ all the processes which are waiting for any I/O device, are kept in it's device queue.
-

In system ⇒ one job queue
one ready queue
> 1 device queue (one for each device)

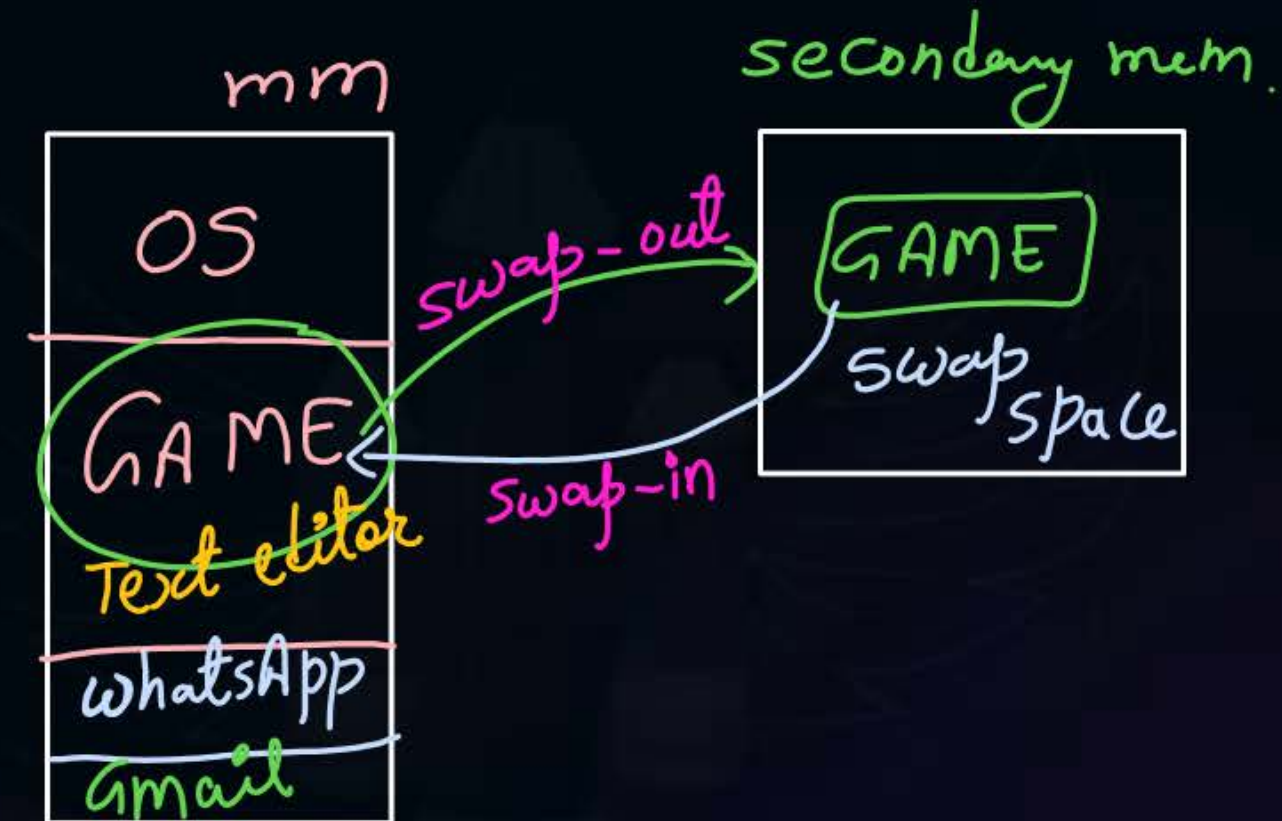


Topic : Types of Schedulers

- Long-Term Scheduler (Job) → selects which process to be taken from new to ready state; and then bring them also.
- Short-Term Scheduler (CPU) → selects one of the ready processes to run next on CPU.
- Mid-Term Scheduler (Medium-term)

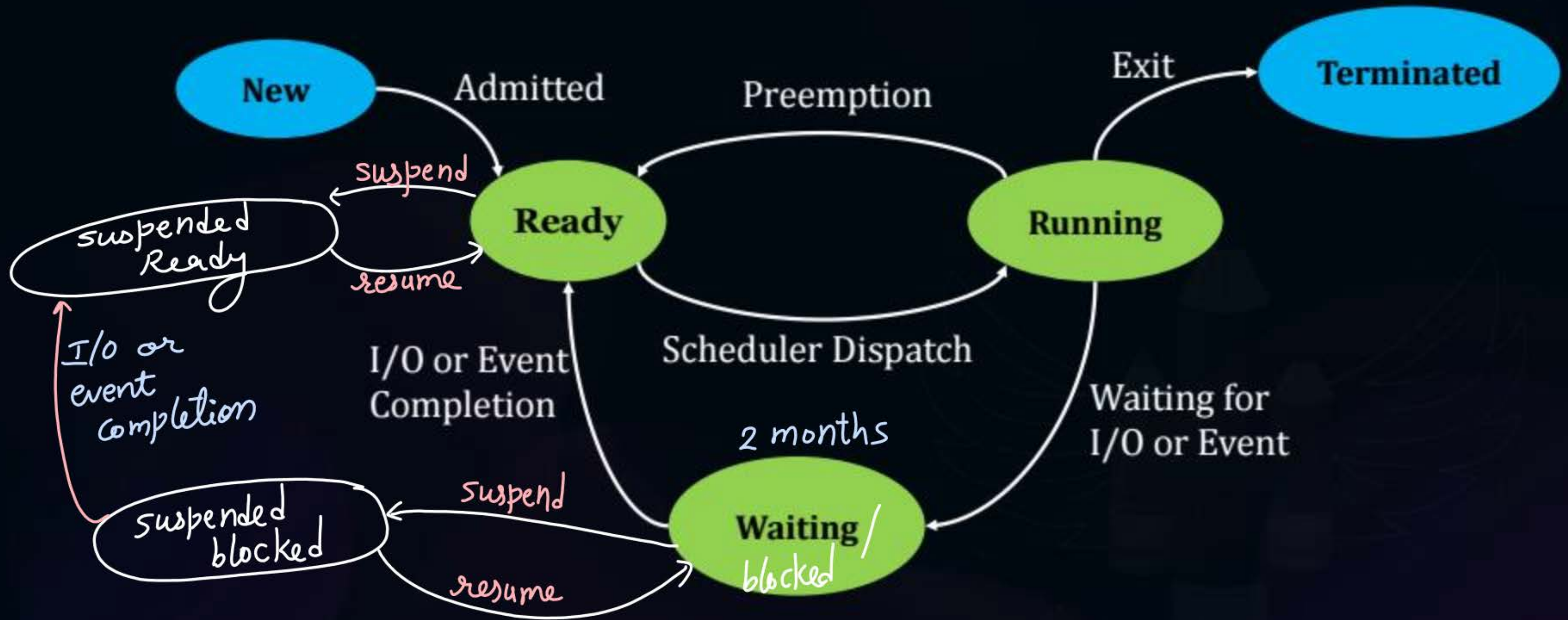
↓
performs swapping

if swap-out swap-in done based on priority of process, then it is called as roll-out and roll-in





Topic : Updated Process States





Topic : CPU Scheduling

Function:

- Make a selection

Goal

- Minimize Wait time and Turn-around time Maximize CPU utilization
- (Throughput) Fairness

[MCQ]

#Q. Which of the following scheduler reduces the degree of multiprogramming?

- A** Short-Term → does not affect degree of multiprogramming
- B** Long-Term → can increase
- C** ✓ Mid-Term → can increase and also can decrease
- D** Long-Term and Mid-Term both



2 mins Summary

Topic

Process states

Topic

Process State Transition

Topic

Process Scheduling

Topic

Types of Schedulers



Happy Learning

THANK - YOU